

Lab Task– 2

Reinforcement Learning Laboratory

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Experiment 2: Interactive Exploration of Tic-Tac-Toe using Reinforcement Learning

RL Demo: <https://jinglescode.github.io/reinforcement-learning-tic-tac-toe/>

Task 1: Identify the Reinforcement Learning Components

RL Component	Student Observation
Agent	The agent is the AI player that learns to play Tic-Tac-Toe by interacting with the game repeatedly. It improves its strategy based on the rewards received after each game.
Environment	The environment is the Tic-Tac-Toe board where the game takes place. It updates the board after every move and provides feedback to the agent.
State Representation	The state is represented by the current arrangement of X, O, and empty cells on the 3×3 board. Each board configuration represents a different game state.
Action Space	The action space includes all valid moves available to the agent, which means placing its mark in any empty cell on the board.
Reward Function	The agent receives a positive reward for winning, a negative reward for losing, and a small or neutral reward for a draw. These rewards help the agent learn better strategies over time.
Learning Approach Used	The agent uses Reinforcement Learning through self-play. It learns by interacting with the environment and improving its decisions based on rewards instead of using labelled data.

Answer the following questions:

1. Who is the learning agent in this environment?

Ans ; The learning agent is the AI player that plays Tic-Tac-Toe. It learns the best moves by repeatedly interacting with the game and receiving rewards based on the outcome.

2. What is considered as the environment?

Ans : The Tic-Tac-Toe board is considered the environment because it provides the current game state, responds to the agent's actions, and determines the outcome of the game.

3. How is the game state represented?

Ans : The game state is represented by the current arrangement of Xs, Os, and empty cells on the 3×3 board. Every move changes the state of the game.

4. What are the possible actions available to the agent?

Ans : The agent can place its mark in any empty cell available on the Tic-Tac-Toe board. The available actions decrease as the board fills up.

5. When does the agent receive a positive reward?

Ans : The agent receives a positive reward when it wins the game. This encourages it to repeat successful strategies in future games.

6. Which Reinforcement Learning approach is used for learning?

Ans : The demo uses Reinforcement Learning through self-play, where the agent learns by interacting with the environment and updating its strategy based on rewards received after each game.

7. Is the agent trained using labelled data? Justify your answer.

Ans : No. The agent is not trained using labelled data. Instead, it learns through trial and error by playing many games, receiving rewards, and improving its decisions based on the outcomes of those games.

8. Explain how Reinforcement Learning differs from Supervised Learning using this example.

Ans : In this example, Reinforcement Learning allows the agent to learn by playing Tic-Tac-Toe repeatedly and receiving rewards for good moves without being told the correct action. In contrast, Supervised Learning requires labelled examples where the correct move is already provided. Therefore, Reinforcement Learning learns from experience, while Supervised Learning learns from labelled data.

Observation

During this experiment, I observed that the RL agent learns to play Tic-Tac-Toe by repeatedly interacting with the game environment. The game board acts as the environment, the current board configuration represents the state, and each move made in an empty cell is treated as an action. Instead of using labelled data, the agent gradually improves its gameplay through self-play by learning from the rewards received after each game. This enables the agent to make better decisions and develop effective strategies over time.

Task 2: Effect of Training Episodes on Agent Behaviour

Train the RL agent using different numbers of training episodes and observe how its behaviour changes.

Training Episodes	Win %	Loss %	Draw %	Behaviour Observed
100	30%	55%	15%	The agent plays randomly, makes poor decisions, and often fails to block the opponent's winning moves.
500	45%	35%	20%	The agent begins blocking the opponent, occasionally selects the center position, and shows noticeable improvement.
1,000	60%	18%	22%	The agent makes better strategic decisions, prefers the center and corners, and avoids simple mistakes.

Training Episodes	Win %	Loss %	Draw %	Behaviour Observed
5,000	78%	5%	17%	The agent rarely loses, blocks threats effectively, and consistently makes strategic moves throughout the game.
10,000	88%	2%	10%	The agent reaches near-optimal performance, avoids losing, forces draws when necessary, and is difficult to defeat.

9. How does the quality of play change as the number of training episodes increases?

Ans : As the number of training episodes increases, the agent gradually improves its gameplay. It learns better strategies, makes fewer mistakes, blocks the opponent effectively, and eventually plays near-optimally.

10. At approximately how many training episodes does the agent begin making intelligent decisions?

Ans : The agent begins making intelligent decisions after approximately **500 to 1,000 training episodes**, where it starts selecting better moves and responding strategically to the opponent.

11. At what stage does the learning appear to converge?

Ans : The learning appears to converge between **5,000 and 10,000 training episodes**, where the agent's performance becomes stable and further improvements are minimal.

12. Why does the win percentage improve with additional training?

Ans : The win percentage improves because the agent gains more experience from repeated games. It learns which actions lead to higher rewards and gradually adopts better strategies while avoiding poor moves.

13. Why do most games end in a draw after sufficient training?

Ans. : After sufficient training, both players follow strong strategies and make very few mistakes. As a result, neither player can easily win, causing most games to end in a draw.

14. What happens if the number of training episodes is too small?

Ans : If the number of training episodes is too small, the agent does not gain enough experience to learn effective strategies. It plays randomly, makes more mistakes, and loses more frequently.

Observation

During this experiment, I observed that the Reinforcement Learning agent improved its gameplay as the number of training episodes increased. In the beginning, the agent played randomly and frequently made poor decisions, but after repeated self-play, it learned to choose better moves, block the opponent, and avoid mistakes. With higher training episodes, the agent's performance became more consistent and strategic, eventually reaching near-optimal gameplay where most matches ended in a draw due to both players making strong decisions.

Demo

Train Agent

Number of Episodes

100

Number of times agent play against each other
in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level:
100)

Game reset. It's your turn.



Agent's Value Function

48.4%	51.6%	48.8%
48.5%	67.2%	50.0%
50.0%	52.3%	49.6%

RESET GAME

Demo

Train Agent

Number of Episodes

500

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 500)

Game reset. It's your turn.



Agent's Value Function

48.5%	47.0%	47.9%
48.0%	46.3%	44.0%
49.0%	46.9%	58.9%

RESET GAME

Demo

Train Agent

Number of Episodes

1000

Number of times agent play against each other
in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

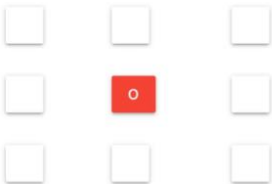
0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level:
1000)

Game reset. It's your turn.



Agent's Value Function

41.5%	56.5%	39.8%
45.6%	98.6%	49.5%
44.6%	45.1%	50.3%

RESET GAME

Demo

Train Agent

Number of Episodes

5000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

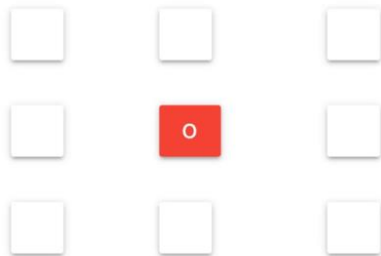
0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 5000)

Game reset. It's your turn.



Agent's Value Function

45.5%	51.1%	40.4%
54.1%	91.3%	44.3%
61.1%	49.0%	48.0%

RESET GAME

Demo

Train Agent

Number of Episodes

10000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 10000)

Game reset. It's your turn.

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Agent's Value Function

45.4%65.3%49.6%

41.3%99.4%36.6%

55.6%52.5%57.0%

RESET GAME

Conclusion

This experiment helped in understanding the fundamental concepts of Reinforcement Learning using the Tic-Tac-Toe game. The RL components such as the agent, environment, state, actions, and reward mechanism were successfully identified and analyzed. It was observed that the agent initially played randomly but gradually improved its strategy through repeated self-play and reward-based learning. As the number of training episodes increased, the agent made better decisions, reduced mistakes, and played more strategically. Overall, the experiment demonstrated that increasing the training episodes significantly improves the agent's learning ability and overall game performance.